

The fewest pieces

Name: _____

Date: _____

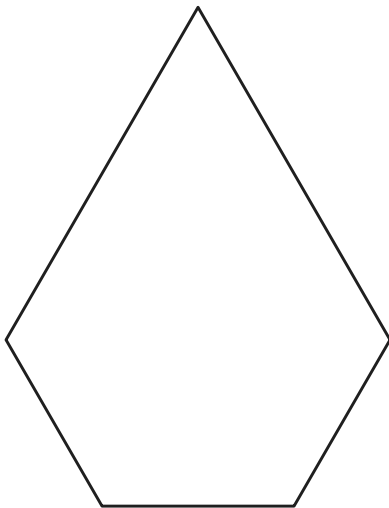
Use only green triangles, blue rhombi, red trapezoids, and yellow hexagons.

1. The pennant / P1

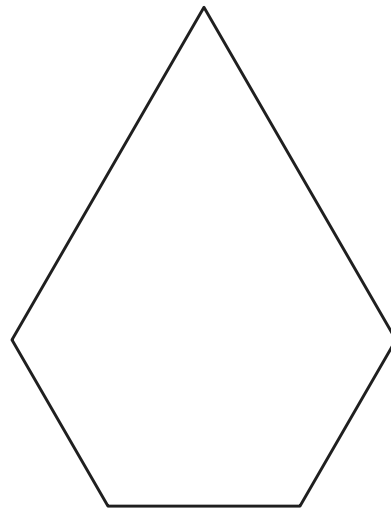
Make a legal pair. Then try to use fewer pieces **in total across both copies**.

Keep both copies built. A color used in A cannot be used in B.
Cover each outline exactly: no gaps, overlaps, or pieces outside.

Copy A



Copy B



Pieces in A: _____ Pieces in B: _____ Total: _____

Could a pair use fewer pieces? Explain with blocks, a drawing, or words.

Think about: How much space could one piece cover? What would a two-piece filling have to use?



Adult: print at **100% / Actual Size**, single-sided. This line is 1 inch (25.4 mm) in the default file. Match a green triangle's edge to it before using the mats.

A bigger shape, a new limit

Name: _____

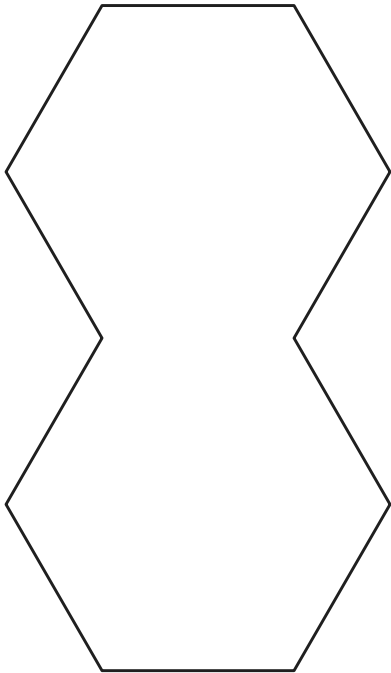
Date: _____

2. Double hexagon / P2

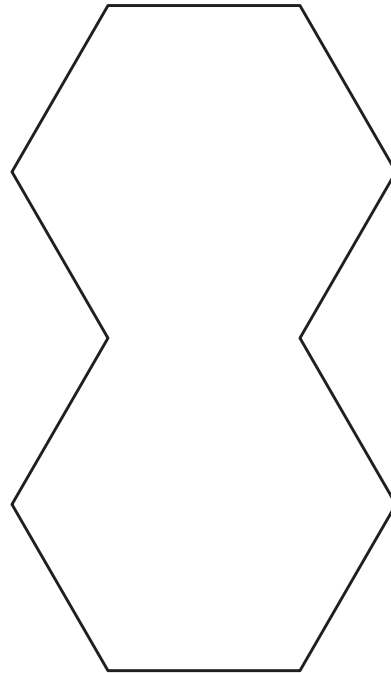
Find the smallest total number of pieces for a legal pair.

Keep both copies built. A color used in A cannot be used in B.
Cover each outline exactly: no gaps, overlaps, or pieces outside.

Copy A



Copy B



Pieces in A: _____ Pieces in B: _____ Total: _____

My best total is _____. Why can no smaller total work?

Think about: If one copy uses yellow, what is the largest block the other copy can use? What if neither uses yellow?

Build first. You may explain by pointing, talking, or drawing.

No yellow. How few now?

Name: _____

Date: _____

3. Double hexagon / P2, no yellow

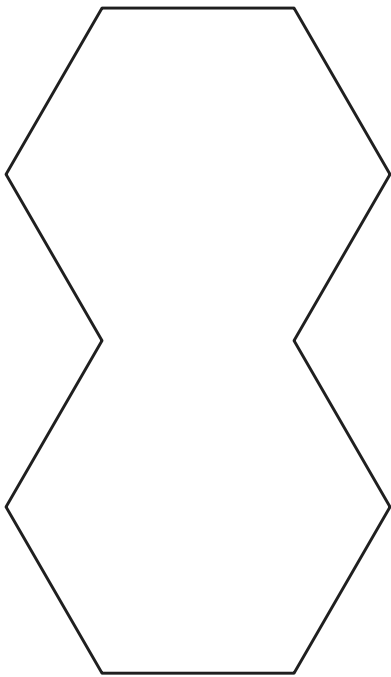
Set yellow aside. Find the smallest total for a legal pair.

Use only green, blue, and red blocks.

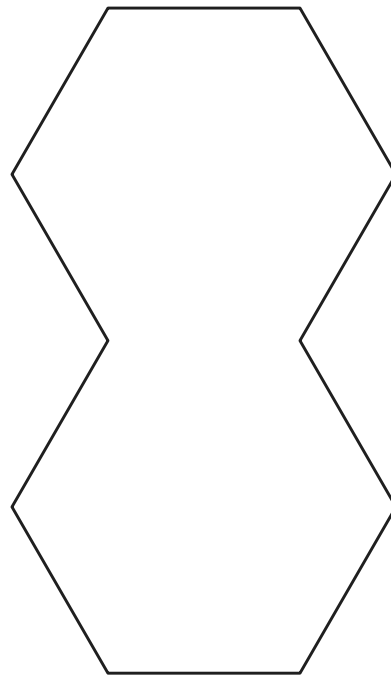
Keep both copies built. A color used in A cannot be used in B.

Cover each outline exactly: no gaps, overlaps, or pieces outside.

Copy A



Copy B



Pieces in A: _____ Pieces in B: _____ Total: _____

Build a pair that works. Explain why a smaller total cannot work.

Think about: Can both copies use red? Does the right total area always mean the pieces will fit?

Build first. You may explain by pointing, talking, or drawing.